Mafatlal M. Kher, PhD

Plant biotechnology · Plant virology · Plant tissue culture

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Professional profile

Plant biotechnologist and plant virologist with more than 15 years of research experience in plant tissue culture, grapevine virology, molecular diagnostics and conservation propagation. Established complete micropropagation protocols for sweet potato and grapevine, from culture initiation and multiplication through rooting and acclimatization; corresponding manuscripts are in preparation. Research interests include virus elimination, somatic embryogenesis and seed biology. Experience spans university teaching, student mentoring, grant writing, editorial service and commercial tissue culture consultancy.

Research and academic appointments

Postdoctoral Researcher | Ariel University, Israel

January 2022-present · Grape and Wine Research Group · Supervisor: Prof. Elyashiv Drori

Develop grapevine propagation and virus-indexing workflows spanning shoot-tip establishment, multiplication, rooting and acclimatization of indigenous varieties. Evaluate chemotherapy and etiolated shoot-tip culture for grapevine virus elimination, with molecular indexing to assess plant material. Perform RNA extraction from leaf and woody tissues, cDNA synthesis, gradient PCR and RT-qPCR indexing. Conduct OIV-based characterization, powdery mildew assessment and fruit-development documentation.

Work on hawthorn seed biology and in vitro propagation, and on culture establishment, callus and shoot culture, and vegetative propagation of *Cupressus*. Contribute to ongoing somatic-embryogenesis protocol development. Research partner and proposal contributor on three KKL-JNF-funded projects with Prof. Drori. Support MSc research and mentoring, PhD recruitment, procurement and instrument service coordination.

Assistant Professor of Biotechnology | GSFC University, India

7 April 2019-14 December 2021 · School of Biosciences

Delivered more than 20 undergraduate and postgraduate courses and developed e-content for six courses. Taught Plant Tissue Culture, Plant Biotechnology, Virology, Research Methodology, Environmental Biotechnology and Developmental Biology; designed and assessed laboratory sessions.

Supervised MSc dissertations and undergraduate and postgraduate research. Served as Officer-in-charge of the Admission Task Force and Student Placement Committee coordinator; founder member of the Biotechnology Club. Convened the June 2021 international plant tissue culture webinar series, with 21 expert talks and more than 1,500 registrations from 42 countries.

Research Scholar | Sardar Patel University, India

2009-2018 · P.G. Department of Biosciences · Supervisor: Prof. M. Nataraj

Researched micropropagation and regeneration of medicinal plants, acclimatization and culture-facility management. Co-supervised MSc projects. Served as Project Fellow on the UGC-sponsored Hyphaene dichotoma micropropagation project from 1 March 2011 to 31 January 2014.

Adhoc Lecturer in Biology | B.C.J. Science College, Khambhat, India

2008-2009

Taught Morphology and Anatomy of Flowering Plants, and Biology and Diversity of Lower Plants.

Education

PhD in Botany | Sardar Patel University, India

Degree awarded March 2020 · Thesis defended December 2018

In vitro propagation of some medicinally important plants. P.G. Department of Biosciences, Vallabh Vidyanagar. Advisor: Prof. M. Nataraj.

MSc in Botany | Sardar Patel University, India

December 2008

Distribution and developmental biology of floral nectaries of Aegiceras corniculatum (L.) Blanco. Advisor: Prof. J.S.S. Mohan.

BSc in Botany | The Maharaja Sayajirao University of Baroda, India

October 2006

Funded research and proposal development

KKL-JNF, Israel: Research partner and proposal contributor with Prof. Elyashiv Drori on three KKL-JNF-funded projects.

Since February 2026: Tissue Culture Propagation and Drought Tolerance Assessment of *Cupressus dupreziana* and *Hesperocyparis guadalupensis*.

Funded since October 2024: Advancing the Conservation and Management of *Cupressus atlantica* Gaussen through Optimization of Somatic Embryogenesis.

2022-2025: Development of methods for propagating spiny hawthorn and red hawthorn from seeds and cuttings. Project began October 2022.

Earlier funded proposals with Prof. M. Nataraj: Proposal writer for Micropropagation of *Ailanthus excelsa* Roxb., a multipurpose tree (GSBTM, 2017-2020), and Micropropagation of *Hyphaene dichotoma*, a rare and endemic palm (UGC, 2011-2014). Awards were made to the eligible principal investigator.

Consultancy and industry engagement

Commercial tissue culture laboratory planning and setup consultancy: Equipment and consumable requirements, staffing, production schedules, aseptic operations and transfer of plants to nursery conditions.

Prem Sarita Agribiotech, Rajpipla, India | 2016-present. Plant tissue culture consultancy; hybrid engagement during 2016-2020 and online engagement from 2020.

Gujarat Forest Department | 2019. Co-opted member of the Plant Tissue Culture Technical Committee, Silviculture and Forest Utilization Circle, Rajpipla. Evaluated tissue culture service providers for *Dendrocalamus*, *Melia* and *Bambusa*.

GSFC Agrotech | 2019-2020. Contributed manuscript writing as coauthor and corresponding author to the 2021 study on micropropagation of strawberry cultivars Sweet Charlie and Winter Dawn, covering rooting and acclimatization. Experimental design and execution were undertaken by coauthors.

Teaching and mentoring

Teaching: Undergraduate and postgraduate plant biotechnology, tissue culture, virology, research methodology, environmental biotechnology and developmental biology at GSFC University; undergraduate botany subjects at B.C.J. Science College.

Supervision and mentoring: Provided mentoring support for MSc research at Ariel University; primarily supervised MSc dissertations and mentored student research and industrial internships at GSFC University; co-supervised 20 MSc projects during the doctoral period at Sardar Patel University.

Editorial service and awards

Editorial Board Member, *BMC Plant Biology* (Springer Nature). Handling Editor, Tissue Culture and Conservation Biology section, *Vegetos*. Guest Editor, *Horticulturae* special issue on Innovations in Micropropagation of Horticultural Plants.

Peer-review service across more than 30 international journals, including *Plant Cell, Tissue and Organ Culture*, *In Vitro Cellular and Developmental Biology - Plant*, *Plant Methods*, *Scientia Horticulturae* and *Industrial Crops and Products*.

2026: Editorial Contribution Award and Author Service Award, *BMC Plant Biology*, Springer Nature.
2018: Outstanding Reviewer Award, *Industrial Crops and Products*, Elsevier.
2016: Outstanding Reviewer Award, *Scientia Horticulturae*, Elsevier.

Technical expertise

Molecular diagnostics: DNA and RNA isolation from woody and recalcitrant tissues; cDNA synthesis; conventional PCR; gel electrophoresis; SYBR Green and TaqMan RT-qPCR/qPCR; QuantStudio 1, Bio-Rad T100, NanoDrop and Bio-Rad Gel Doc.

Plant tissue culture: Micropropagation; shoot tip and meristem culture; somatic embryogenesis; organogenesis; ex vitro and in vivo rooting; acclimatization; MS and modified media formulation; cytokinin and auxin balancing; contamination control.

Seed biology: Physical and physiological dormancy; chemical and mechanical scarification; stratification; imbibition and germination kinetics; early-seedling biochemistry; viability testing.

Histology and microscopy: Paraffin embedding, sectioning and staining for light microscopy; stereo-zoom microscopy; embryo developmental staging.

Data and research tools: GraphPad Prism, SPSS, R, BioRender and Figure Lab; Mendeley and EndNote; Web of Science, Scopus, PubMed and Google Scholar; literature screening, reference verification and research workflows using large language models.

Laboratory operations: Procurement, inventory and vendor management, instrument service coordination and maintenance follow-up.

Professional development

2025: PLANT Workshop - Plant Sciences in the Anthropocene, Université Paris-Saclay, Institut Pascal, France, 24 March-4 April.

2024: Data Carpentry Genomics training, Weizmann Institute of Science, 29-31 October.

2020: Online Teaching and Assessment Pedagogy, TEQIP-III Faculty Development Programme, Dibrugarh and Panjab University, 24-29 August.

2020: Coursera courses: Introduction to the Biology of Cancer with honours (Johns Hopkins University); The Science of Stem Cells (AMNH); Understanding Plants Parts I and II (Tel Aviv University); Introduction to Algae (UC San Diego); DNA Decoded (McMaster University).

2018: Elsevier Researcher Academy - Publishing Open Access and Funding Hacks for Researchers.
2016: Springer Author Academy - nine modules covering journal and book authorship, peer review, open access and scientific English.

Professional membership and languages

Society for Plant Research / VEGETOS, India (since 2026).

English: professional working proficiency. Gujarati: native. Hindi: advanced. Marwadi and Wagadi: conversational. Sanskrit: intermediate.

Invited talks and keynote lectures

Kher MM (2026). Plant tissue culture and biodiversity conservation. International Conference on Biodiversity Protection: Its role in sustainable development, 18-19 May 2026, Janardan Rai Nagar Rajasthan Vidyapeeth, Udaipur, Rajasthan, India. (Online)

Kher MM (2024). In vitro propagation of bamboo. Invited lecture, Garden City University, Bangalore, India. (Online)

Kher MM (2024). Harnessing millet: biotechnological innovations. Invited talk, international conference, Garden City University, India. (Online)

Kher MM (2023). Strategies for commercial plant tissue culture. Invited talk, Quest International University (Malaysia) and SAGE University (India). (Online)

Kher MM (2023). Innovative solutions for sustainable agriculture through plant tissue culture. Invited talk, Saurashtra University, India. (Online)

Kher MM (2022). How professional memberships can advance your career. Invited talk, BHU Varanasi and Geochemical Society, India. (Online)

Kher MM (2020). Bamboo the green gold. Keynote, International Webinar on Plants for Sustainable Living of Humankind, SPMVV, Tirupati, India. (Online)

Kher MM (2019). Plant tissue culture: problems and solutions. Keynote, National Workshop on Commercial PTC, Parul University, India. (In-person)

Conference presentations

Oral presentations

Etiolated shoot tip culture: a novel approach for virus-free grapevine. Oral, Vine and Wine Research Institute, Ariel University, Israel. 2025

Thidiazuron-induced somatic embryogenesis and organogenesis from leaves and internodes of Pluchea lanceolata. National Conference on Current Trends in Biological Sciences-II, Sardar Patel University, Vallabh Vidyanagar, India, 9 February 2018.

In vitro propagation of Clerodendrum phlomidis Linn. International Conference on Current Status, Opportunities and Challenges in Medicinal Plants and Natural Product Research, Uka Tarsadia University, Bardoli, India, 24 to 26 September 2014.

Micropropagation of Uraria picta. National Conference on Biodiversity and Its Role in the Modern World, SPDM/SSB/SHD/SMA College, Shirpur, India, 12 to 13 October 2012.

Research proposal in biotechnology. National Workshop on Research Methodology, Sardar Patel University, India, 12 to 14 March 2012.

Poster presentations

Kher MM, Drori E (2025). Etiolated shoot tip culture: a novel method for virus-free grapevine propagation. International Advances in Plant Virology, Murcia, Spain, 8-11 April 2025.

Kher MM, Viner L, Shecori S, Rahimi O, Lubin BCR, Drori E (2023). In vitro propagation of ancient indigenous grapevine varieties of Israel. Vine and Wine Research Institute conference, Ariel University, Israel, 16 May 2023.

Micropropagation of Dendrocalamus strictus (Roxb.) Nees for social forestry. National Conference on Current Trends in Biological Sciences-V, Sardar Patel University, India, 15 February 2020.

Source of callus origin and plant growth regulators responsible for multiple regeneration pathways in Crataeva nurvala Buch Ham. International Conference on Biomolecular and Biotechnological Aspects of Medicinal Plants, L.M. College of Pharmacy, Ahmedabad, India, 18 to 19 October 2019.

Micropropagation of Merremia quinquefolia (L.) Hallier F. from nodal explants. National Symposium on Microbial Biotechnology, N.V. Patel Science College, Vallabh Vidyanagar, India, 26 February 2014.

Micropropagation of Hibiscus radiatus, a medicinal plant. Impact of Plant Tissue Culture on Advances in Plant Biology and XXXIII PTCA(I) Annual Meet, Ahmedabad, India, 19 to 21 January 2012.

Callus induction and plant regeneration in Crataeva nurvala (Buch. Ham.). Zonal Seminar, B.A. College of Agriculture, Anand Agricultural University, India, 17 January 2012.

Publications

44 journal articles and reviews, and one book chapter. Full publication record follows, grouped by output type.

Research articles

1. Collince Omondi Awere, Kasinathan Rakkammal, P.S. Jeevan Ram, K. Pavan Kumar, Kaliraj Ragavan, R. Anitha Kumari, Suresh Govindan, **Mafatlal M. Kher**, Elyashiv Drori, Manikandan Ramesh (2024). Micropropagation, genetic fidelity and chromatographic analysis in Evolvulus alsinoides (L.): a potent multipurpose medicinal plant. Industrial Crops and Products, 213, 118444. doi:10.1016/j.indcrop.2024.118444

2. Shani Shecori, **Mafatlal M. Kher**, Aharon Azagury, Elyashiv Drori (2024). The potential of grapevine leaf extract in treating hyperpigmentation. Cosmetics, 11(1), 2. doi:10.3390/cosmetics11010002

3. Oshrit Rahimi, Jotham Ziffer Berger, Ilana Shtein, **Mafatlal M. Kher**, Suembikya Frumin, Sariel Hubner, Ehud Weiss, Elyashiv Drori (2023). Wild grapevine (Vitis vinifera L. subsp. sylvestris (C.C. Gmelin) Hegi): novel species to the Israeli flora. Horticulturae, 9(9), 998. doi:10.3390/horticulturae9090998

4. Shani Shecori, **Mafatlal M. Kher**, Kamal Tyagi, Larry Lerno, Yishai Netzer, Amnon Lichter, Susan E. Ebeler, Elyashiv Drori (2022). A field collection of indigenous grapevines as a valuable repository for applied research. Plants, 11(19), 2563. doi:10.3390/plants11192563

5. Aaqib Shaheen, Yaser Hassan Dewir, **Mafatlal M. Kher**, Majid Khan, Antar Nasr El-Banna, Ahmed Alaizari (2022). Synergistic effect of benzylaminopurine and meta-Topolin combination for micropropagation of gerbera 'Pink Melody'. Ciencia e Agrotecnologia, 46, e017521. doi:10.1590/1413-7054202246017521

6. Madanmohan R. Dhukate, **Mafatlal M. Kher**, A.V. Vadawale, Priyanka Giri (2021). Protocol for micropropagation of strawberry (Fragaria × ananassa) cv. 'Sweet Charlie' and 'Winter Dawn'. Environmental and Experimental Biology, 19, 1-6. doi:10.22364/eeb.19.01

7. **Mafatlal M. Kher**, M. Nataraj (2020). In vitro regeneration competency of Crataeva nurvala (Buch Ham) callus. Vegetos, 33(1), 52-62. doi:10.1007/s42535-019-00080-x

8. Bharat S. Rajput, Minal Jani, Kasuladev Ramesh, M. Manokari, Phanikanth Jogam, Venkateswar Rao Allini, **Mafatlal M. Kher**, Mahipal S. Shekhawat (2020). Large-scale clonal propagation of Bambusa balcooa Roxb.: an industrially important bamboo species. Industrial Crops and Products, 157, 112905. doi:10.1016/j.indcrop.2020.112905

9. Yaser Hassan Dewir, Abdulhakim A. Aldubai, **Mafatlal M. Kher**, Abdullah A. Alsadon, Salah El-Hendawy, Nasser A. Al-Suhaibani (2020). Optimization of media formulation for axillary shoot multiplication of the red-peeled sweet potato (Ipomoea batatas [L.] Lam.) 'Abees'. Chilean Journal of Agricultural Research, 80(1), 3-10. doi:10.4067/S0718-58392020000100003

10. J. Revathi, M. Manokari, R. Latha, S. Priyadharshini, **Mafatlal M. Kher**, Mahipal S. Shekhawat (2020). Photoperiod and silver ions modulate in vitro flowering in Oldenlandia herbacea (L.) Roxb. Israel Journal of Plant Sciences, 67(3-4), 219-224. doi:10.1163/22238980-bja10014

11. **Mafatlal M. Kher**, Dilipsingh B. Rajput, Dipti Damor, Dhaval Khandhar, M. Nataraj (2019). Direct shoot regeneration from excised leaf segments of Crataeva nurvala. Environmental and Experimental Biology, 17, 179-183. doi:10.22364/eeb.17.18

12. **Mafatlal M. Kher**, M. Nataraj (2019). Direct somatic embryogenesis and shoot regeneration from leaves and internodes of Pluchea lanceolata (DC.) C.B. Clarke. In Vitro Cellular & Developmental Biology - Plant, 55(6), 720-724. doi:10.1007/s11627-019-10016-4

13. J. Revathi, M. Manokari, R. Latha, S. Priyadharshini, **Mafatlal M. Kher**, Mahipal S. Shekhawat (2019). In vitro propagation, in vitro flowering, ex vitro root regeneration and foliar micro-morphological analysis of Hedyotis biflora (Linn.) Lam. Vegetos, 32(4), 609-619. doi:10.1007/s42535-019-00066-9

14. **Mafatlal M. Kher**, M. Nataraj (2017). Micropropagation of Combretum ovalifolium Roxb.: a medicinally important plant. Rendiconti Lincei. Scienze Fisiche e Naturali, 28(3), 519-527. doi:10.1007/s12210-017-0625-z

15. Harsh Joshi, Sureshkumar Nekkala, Deepak Soner, **Mafatlal M. Kher**, M. Nataraj (2016). In vitro shoot multiplication of Withania coagulans (Stocks) Dunal. Plant Tissue Culture and Biotechnology, 26(2), 187-195. doi:10.3329/ptcb.v26i2.30569

16. **Mafatlal M. Kher**, Deepak Soner, Neha Srivastava, M. Nataraj, Jaime A. Teixeira da Silva (2016). Micropropagation of Clerodendrum phlomidis L.f. Journal of Horticultural Research, 24(1), 21-28. doi:10.1515/johr-2016-0003

17. Anjali Bose, **Mafatlal M. Kher**, M. Nataraj, Haresh Keharia (2016). Phytostimulatory effect of indole-3-acetic acid by Enterobacter cloacae SN19 isolated from Teramnus labialis (L. f.) Spreng rhizosphere. Biocatalysis and Agricultural Biotechnology, 6, 128-137. doi:10.1016/j.bcab.2016.03.005

18. **Mafatlal M. Kher**, M. Nataraj (2015). Effect of sulfuric acid treatment on breaking of seed dormancy and germination of Indian doum palm, Hyphaene dichotoma, a threatened and endemic palm. Environmental and Experimental Biology, 13, 99-101.

19. **Mafatlal M. Kher**, M. Nataraj, Hettal D. Parmar, Hasmatbanu Buchad (2015). Micropropagation of Merremia quinquefolia (L.) Hallier F. from nodal explants. Journal of Horticultural Research, 23(1), 13-16. doi:10.2478/johr-2015-0002

20. Deepak Soner, M. Nataraj, **Mafatlal M. Kher** (2015). Somatic embryogenesis in Clerodendrum phlomidis L. Academic Journal of Plant Sciences, 7(1), 6-9. doi:10.5829/idosi.ajps.2015.7.1.9267

21. Bhavin Suthar, Jayesh Pansuriya, **Mafatlal M. Kher**, Vinay R. Patel, M. Nataraj (2014). Biochemical changes under chromium stress on germinating seedlings of Vigna radiata. Notulae Scientia Biologicae, 6(1), 77-81. doi:10.15835/nsb.6.1.9203

22. **Mafatlal M. Kher**, Dimpal Joshi, Sureshkumar Nekkala, M. Nataraj, Dharmesh P. Raykundaliya (2014). Micropropagation of Pluchea lanceolata (Oliver & Hiern.) using nodal explant. Journal of Horticultural Research, 22(1), 35-39. doi:10.2478/johr-2014-0004

23. **Mafatlal M. Kher**, M. Nataraj, A. Joshi, M. Patel (2010). Nuptial nectaries in some species of Bignoniaceae. Prajna - Journal of Pure and Applied Sciences, 18, 31-33.

24. M. Nataraj, A. Parmar, **Mafatlal M. Kher** (2009). Impact of chromium during early germination of fenugreek in interaction with copper, zinc, lead, and cadmium. Prajna - Journal of Pure and Applied Sciences, 17, 40-47.

Reviews and mini-reviews

1. Dhaval Patel, Jyoti Solanki, **Mafatlal M. Kher**, Aharon Azagury (2024). A review: surface engineering of lipid-based drug delivery systems. Small, 20(37), 2401990. doi:10.1002/smll.202401990

2. Rupa Sanyal, Saheli Nandi, Sharmila Pandey, Tuyelee Das, Prabhjot Kaur, Merinashwari Konjengbam, Nishi Kant, Md. Habibur Rahman, Avinash Mundhra, **Mafatlal M. Kher**, Uttpal Anand, Radha, Manoj Kumar, Niraj Kumar Jha, Saurabh Kumar Jha, Mahipal S. Shekhawat, Devendra Kumar Pandey, Abhijit Dey (2022). In vitro propagation and secondary metabolite production in Gloriosa superba L. Applied Microbiology and Biotechnology, 106(16), 5399-5414. doi:10.1007/s00253-022-12094-8

3. A.N. Arunkumar, Rekha Ravindranath Warrier, **Mafatlal M. Kher**, Jaime A. Teixeira da Silva (2022). Indian rosewood (Dalbergia latifolia Roxb.): biology, utilization, and conservation practices. Trees - Structure and Function, 36(3), 883-898. doi:10.1007/s00468-021-02243-3

4. **Mafatlal M. Kher**, M. Nataraj, A.N. Arun Kumar, Viji Sitther, Mahipal S. Shekhawat, Rekha R. Warrier, Jaime A. Teixeira da Silva (2021). Tissue culture of Indian rosewood (Dalbergia latifolia Roxb.). Biologia, 76(12), 3595-3604. doi:10.1007/s11756-021-00914-7

5. **Mafatlal M. Kher**, Mahipal S. Shekhawat, M. Nataraj, Jaime A. Teixeira da Silva (2020). Indian sarsaparilla, Hemidesmus indicus (L.) R. Br. ex Schult: tissue culture studies. Applied Microbiology and Biotechnology, 104(15), 6463-6479. doi:10.1007/s00253-020-10714-9

6. Jaime A. Teixeira da Silva, Esmaeil Nezami-Alanagh, Maria E. Barreal, **Mafatlal M. Kher**, Adhityo Wicaksono, Andrea Gulyas, Norbert Hidvegi, Katalin Magyar-Tabori, Nora Mendler-Drienyovszki, Laszlo Marton, Mariana Landin, Pedro Pablo Gallego, John A. Driver, Judit Dobranszki (2020). Shoot tip necrosis of in vitro plant cultures: a reappraisal of possible causes and solutions. Planta, 252(3), 47. doi:10.1007/s00425-020-03449-4

7. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, M. Nataraj (2019). Red sandalwood (Pterocarpus santalinus L. f.): biology, importance, propagation and micropropagation. Journal of Forestry Research, 30(3), 745-754. doi:10.1007/s11676-018-0714-6

8. Dinesh Vasava, **Mafatlal M. Kher**, M. Nataraj, Jaime A. Teixeira da Silva (2018). Bael tree (Aegle marmelos (L.) Correa): importance, biology, propagation, and future perspectives. Trees - Structure and Function, 32(5), 1165-1198. doi:10.1007/s00468-018-1754-4

9. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, M. Nataraj (2018). Indian kino tree (Pterocarpus marsupium): propagation, micropropagation, and biotechnology. Environmental and Experimental Biology, 16, 1-8. doi:10.22364/eeb.16.01

10. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, M. Nataraj, Judit Dobranszki, Melissa A. Millar (2018). Santalum molecular biology: molecular markers for genetic diversity, phylogenetics and taxonomy, and genetic transformation. Agroforestry Systems, 92(5), 1301-1315. doi:10.1007/s10457-017-0075-8

11. Wen-Lu Bi, Chen Pan, Xin-Yi Hao, Zhen-Hua Cui, **Mafatlal M. Kher**, Zvjezdana Markovic, Qiao-Chun Wang, Jaime A. Teixeira da Silva (2017). Cryopreservation of grapevine (Vitis spp.): a review. In Vitro Cellular & Developmental Biology - Plant, 53(5), 449-460. doi:10.1007/s11627-017-9822-9

12. Jaime A. Teixeira da Silva, Songjun Zeng, Adhityo Wicaksono, **Mafatlal M. Kher**, Haenghoon Kim, Munetaka Hosokawa, Yaser Hassan Dewir (2017). In vitro propagation of African violet: a review. South African Journal of Botany, 112, 501-507. doi:10.1016/j.sajb.2017.05.018

13. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, Murugan Nataraj (2016). African sandalwood or Nepalese sandalwood: a brief synthesis. Notulae Scientiae Biologicae, 8(1), 57-61. doi:10.15835/nsb.8.1.9714

14. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, M. Nataraj (2016). Biotechnological advances in Vitex species, and future perspectives. Journal of Genetic Engineering and Biotechnology, 14(2), 335-348. doi:10.1016/j.jgeb.2016.09.004

15. M. Nataraj, **Mafatlal M. Kher**, Jaime A. Teixeira da Silva (2016). Micropropagation of Clerodendrum L. species: a review. Rendiconti Lincei. Scienze Fisiche e Naturali, 27(2), 169-179. doi:10.1007/s12210-015-0484-4

16. **Mafatlal M. Kher**, M. Nataraj, Jaime A. Teixeira da Silva (2016). Micropropagation of Crataeva L. species. Rendiconti Lincei. Scienze Fisiche e Naturali, 27(1), 157-167. doi:10.1007/s12210-015-0478-2

17. Jaime A. Teixeira da Silva, Yaser Hassan Dewir, Adhityo Wicaksono, **Mafatlal M. Kher**, Haenghoon Kim, Munetaka Hosokawa, Songjun Zeng (2016). Morphogenesis and developmental biology of African violet (Saintpaulia ionantha H. Wendl.). Journal of Plant Development, 23, 15-25.

18. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, M. Nataraj (2016). Sandalwood spike disease: a brief synthesis. Environmental and Experimental Biology, 14, 199-204. doi:10.22364/eeb.14.26

19. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, Tony Page, Xinhua Zhang, M. Nataraj, Guohua Ma (2016). Sandalwood: basic biology, tissue culture, and genetic transformation. Planta, 243(4), 847-887. doi:10.1007/s00425-015-2452-8

20. Jaime A. Teixeira da Silva, **Mafatlal M. Kher**, Deepak Soner, M. Nataraj (2015). Withania coagulans (Stocks) Dunal: biotechnological achievements and perspectives. Journal of Horticultural Research, 23(1), 5-12. doi:10.2478/johr-2015-0001

Book chapter

1. **Mafatlal M. Kher**, M. Nataraj (2012). Micropropagation of Hibiscus radiatus: a medicinal plant. In: Braganza, V. (Ed.), Impact of Plant Tissue Culture on Advances in Plant Biology. Loyola Centre for Research and Development, St. Xavier's College, Ahmedabad, India, pp. 18-26.